Building Intelligent, Event-Driven Architectures with Fabric Real-Time Intelligence



Johan Ludvig Brattås, Brian Bønk

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WIFI: FABCON26 – PWD: fabcon-26!

Johan Ludvig Brattås



Lives in Norway

Head of Data & AI at Enora and
Microsoft Data Platform MVP

- specializes in data platform architectures
- works with data in all shapes and sizes
- chronically curious

Brian Bønk

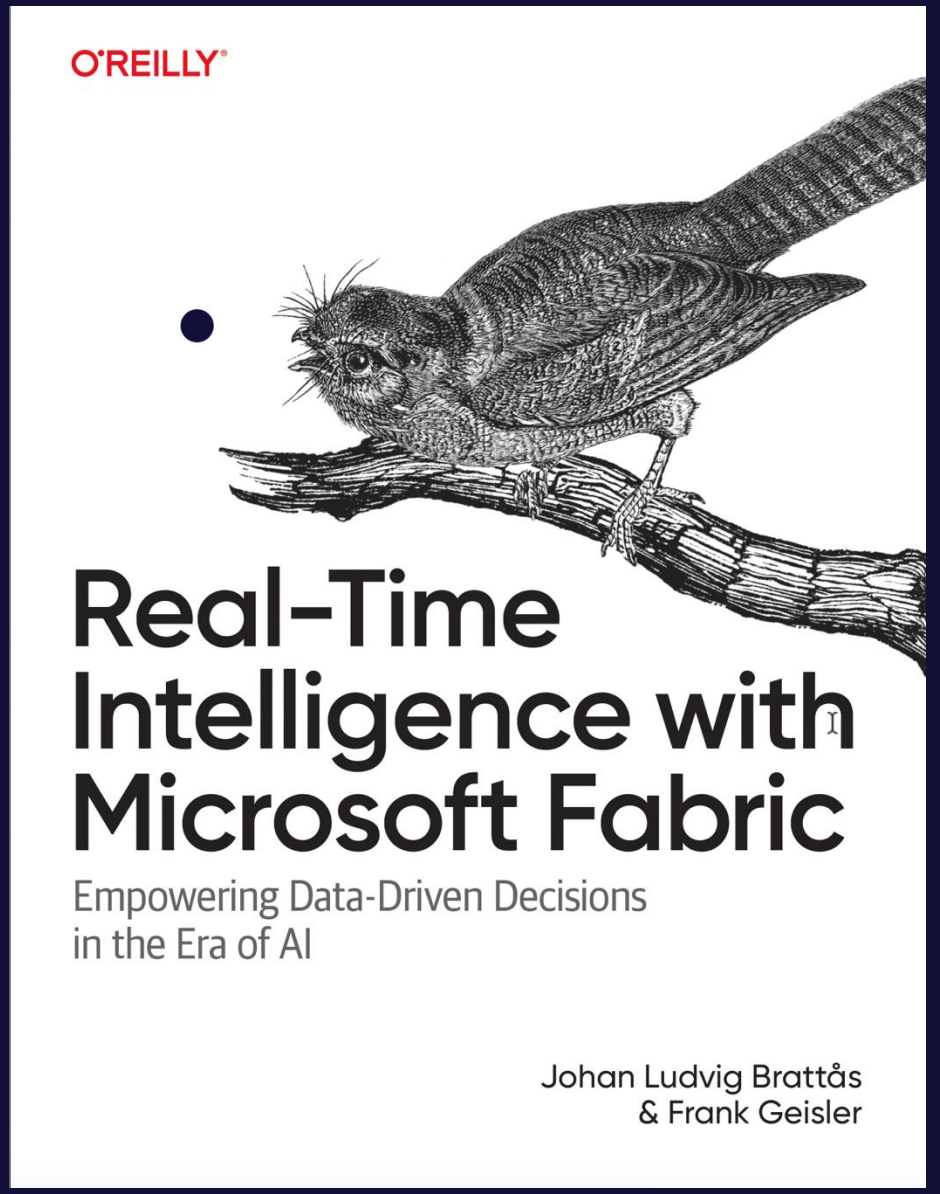


Lives in Denmark

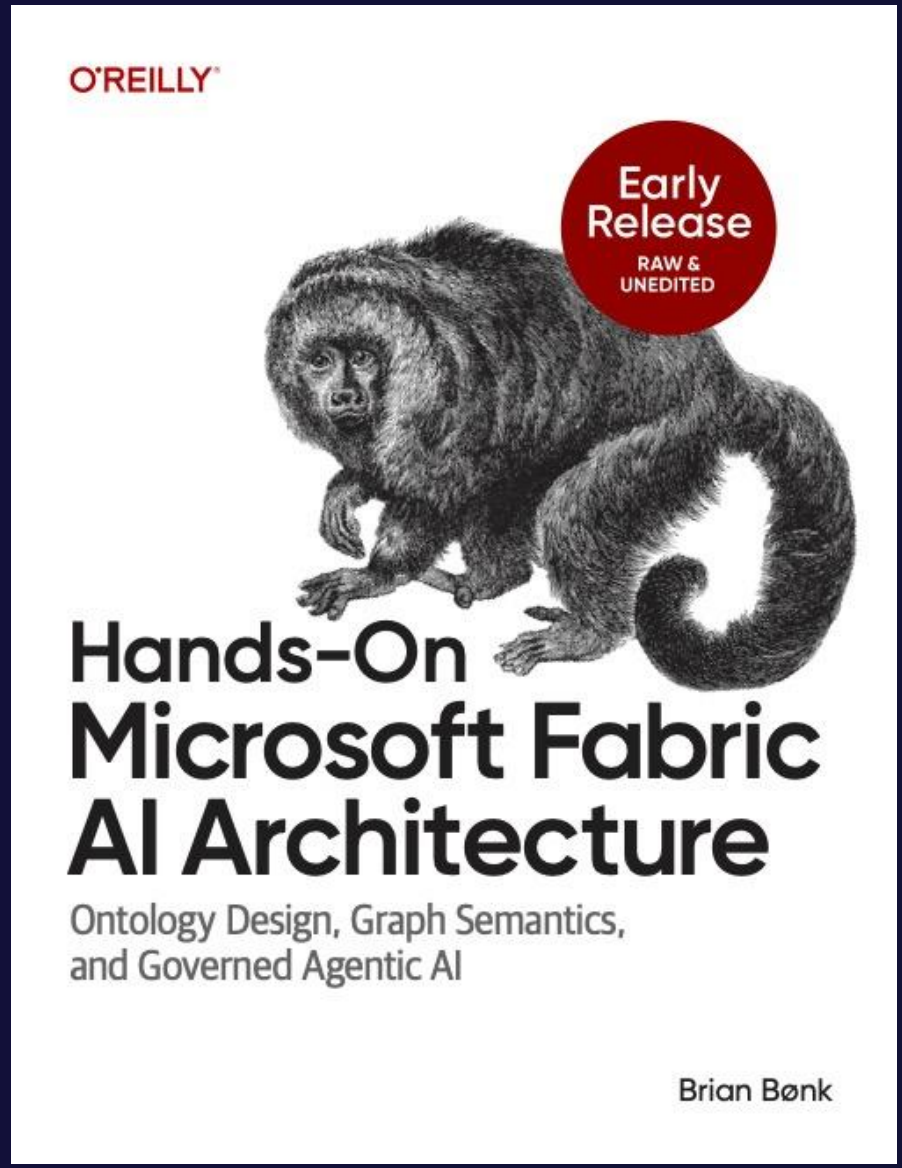
CEO - ProBI and a Microsoft Data Platform MVP

- Real-Time Intelligence
- Analytics Engineering
- Architecture.

Johan & Frank's book on Real-Time Intelligence in Fabric is out now.



Brian’s book on Fabric IQ is soon out! Available as Early access on O’Reilly now.



What can you expect today?

- Dive into event-driven architectures and product capabilities
- Use cases and examples
- Hands-on lab

What we expect from you?

- Be curious, ask questions (raise your hand)
- Follow the instructions for hands-on lab
- Be kind to your neighbours, help each other to learn

Today’s agenda

09:00 – 10:30	Welcome and introduction Event-driven Architectures – Why & What? Setting up the lab environment
10:30 – 11:00	Morning Break
11:00 – 12:45	Hands-on lab - RTI Product Deep Dives
12:45 – 14:00	Lunch Break
14:00 – 15:15	Fabric IQ Continuing the Product Deep Dives
15:15 – 15:45	Afternoon Break
15:45 – 17:00	Hands-on lab – Fabric IQ

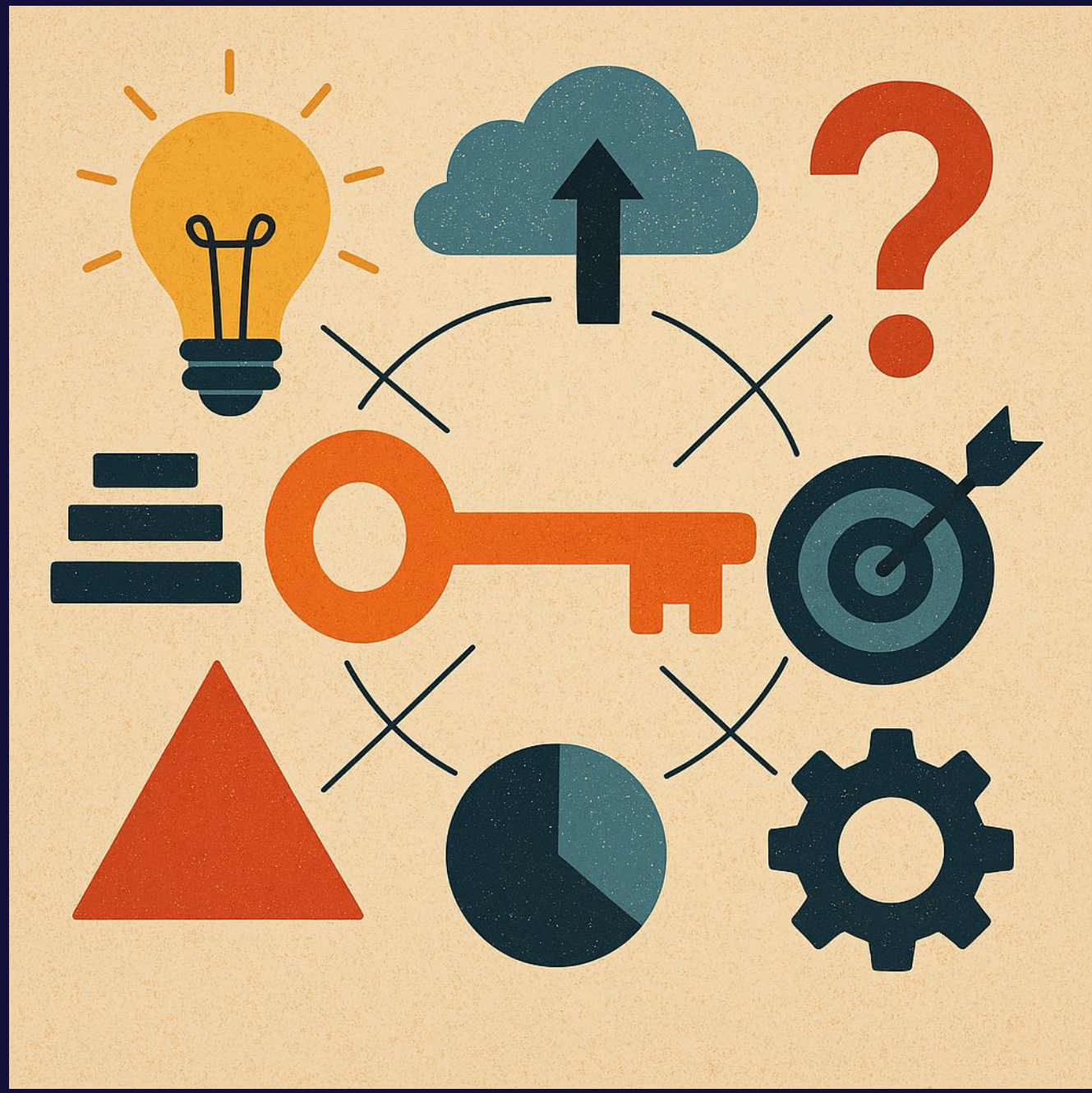
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Event-driven Architectures and Patterns

Key Terminology

- Message: Generic payload of information from one system to another
- Event: A significant change of state
- Event Notification: Often called an «event» but is the immutable, historic message about an event
- Command: A message from one system to another with information on action to be taken



EVENT-DRIVEN ARCHITECTURE



Event-driven architecture is not just for analytics

Three main components:

- Producer
- Event Channel
- Consumer

Asynchronous, decoupled communication between systems.

Central enabler of microservices and other distributed systems

Event-Driven Architecture

When to use Event-driven architecture

- Multiple systems needs the same events
- Real-time processing
- Complex event processing
- High volume + high velocity
- Need for decoupling for scalability and reliability

- Have analytical workloads that don’t fit in traditional scheduling



Common business and operational needs today

- I want to know when something changes in my operational systems (database)
- I want continuous, granular insights from my assets
- I want near real-time reporting, ML scoring and GenAI apps
- I want to process/act/react on events occurring in my physical or digital asset
- I want real-time view into customer interactions and transactions

Event Services in Fabric



Eventhouse / KQL DB – powerful event store. Fits both event sourcing and event-driven architectures. Can be used for simple event processing and analytical loads as well as event stream processing…



Eventstream – Swiss Army knife… Works both for PUB/SUB and event streaming. And for most variations



Data Activator – takes action. Triggers on specified commands or events. Low code/No code

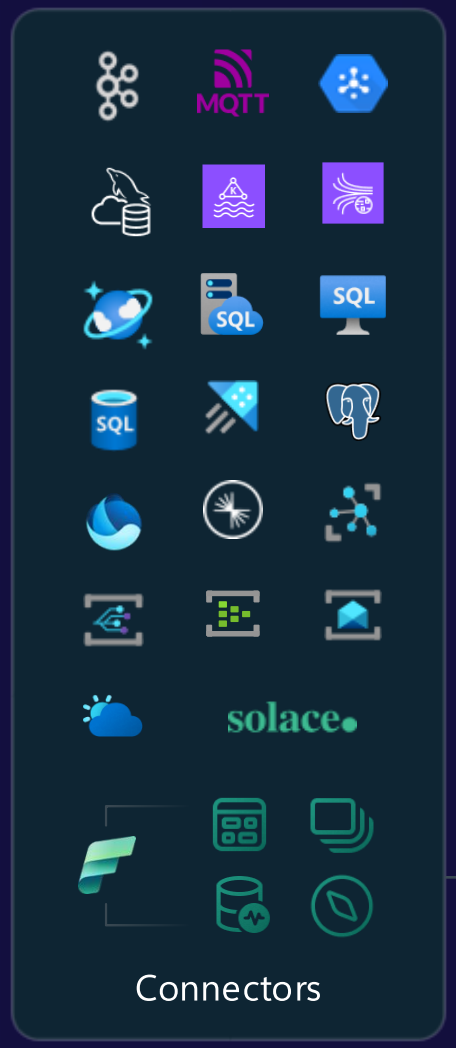
What services should I use?

For Fabric the services rarely overlap.

- Do I have an external event channel?
- Which consumer variation patterns do I have?
- Do I need to take action on events?
- Do I need an event store?

- If no external event channel, complex or event stream scenarios – then Eventstream has a place
- If external event channel and need for an event store KQL DB can be a good fit
- Taking actions on events – such as triggering a pipeline then Activator

Real-Time Intelligence in Microsoft Fabric



Stream Processing

Real-Time Analytics

Digital Twin Models

Real-Time Dashboards

Event-Driven Rules

Actions

Fabric Platform

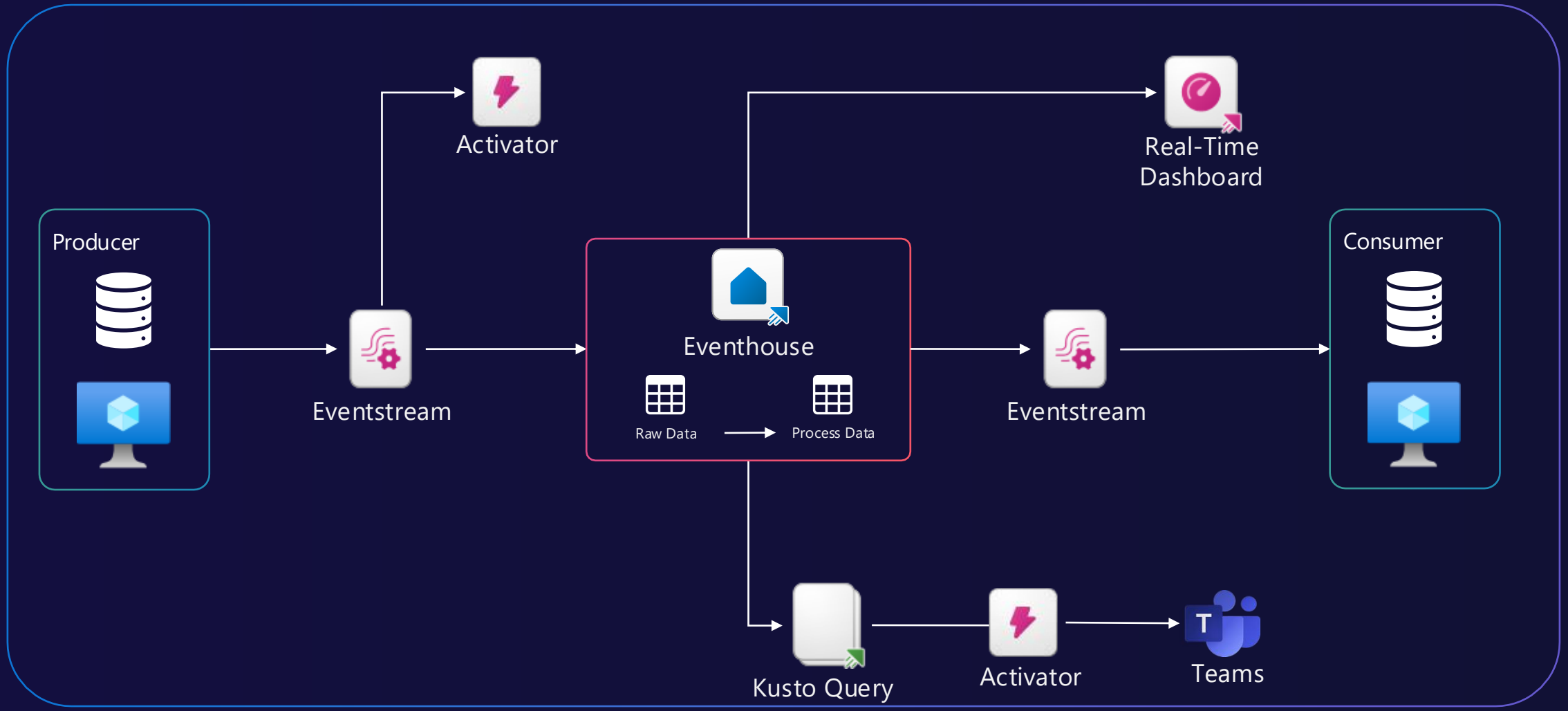
AI

OneLake

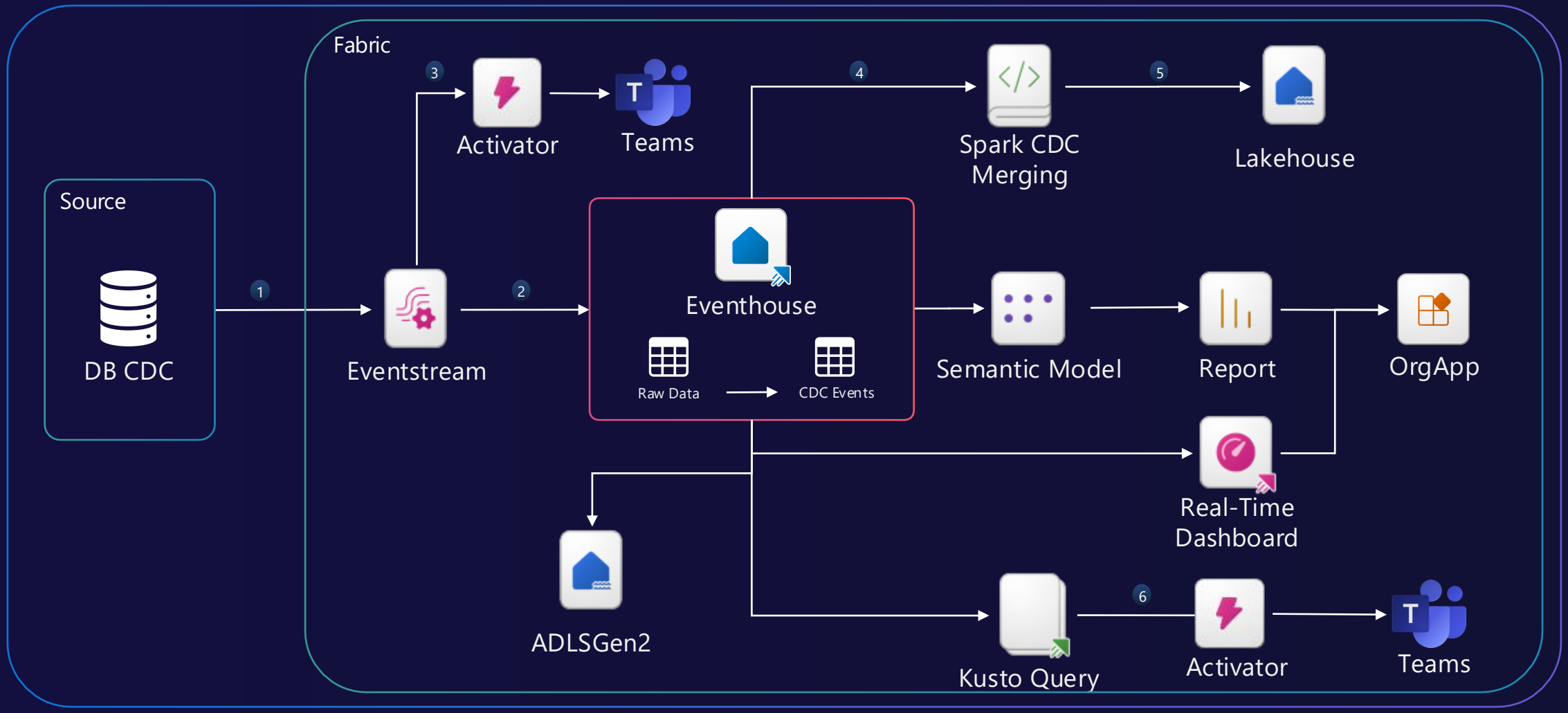
Real-Time Hub

Governance

Pub-Sub Stream Processor



Database Change Data Capture Processing



Setting up the lab

Scenario

- Everything in this section runs on one connecting narrative: **live Barcelona transit around the venue**. An Azure Function polls Transports Metropolitans de Barcelona (TMB) for real-time bus arrival predictions (iBus) and metro arrivals at a curated set of stops and stations around the CCIB and the city's main interchanges, and publishes **each raw API response as an envelope** to Azure Event Hubs.

- You get to ingest that stream with Eventstream, **flatten the nested JSON** into per-bus events, enrich it in an Eventhouse, visualize it on a Real-Time Dashboard, detect conditions with Activator, and then broaden the picture to *non-telemetry* events (OneLake, Fabric job and workspace events) to see the same event-driven pattern applied to data-platform automation.

- The scenario is deliberately different from the afternoon's retail cold-chain scenario.

- Attendees learn the RTI building blocks on transit data in the morning; in the afternoon they see the same blocks re-used (FreezerTelemetryRaw → FreezerTelemetryEnriched → Activator) underneath Fabric IQ's ontology and agents.

Morning section

- Module 1: RTI landscape – basic overview

- Module 2: Eventstream
- Module 3: KQL
- Module 4: Real-time dashboards
- Module 5: Activator
- Module 6: Fabric Events

Instructions

- Use a different browser or private tab to avoid using your existing credentials
- Navigate to https://app.fabric.Microsoft.com
- Login with your username and password available on your desk
- Skip MFA setup

Login instructions

- app.fabric.Microsoft.com

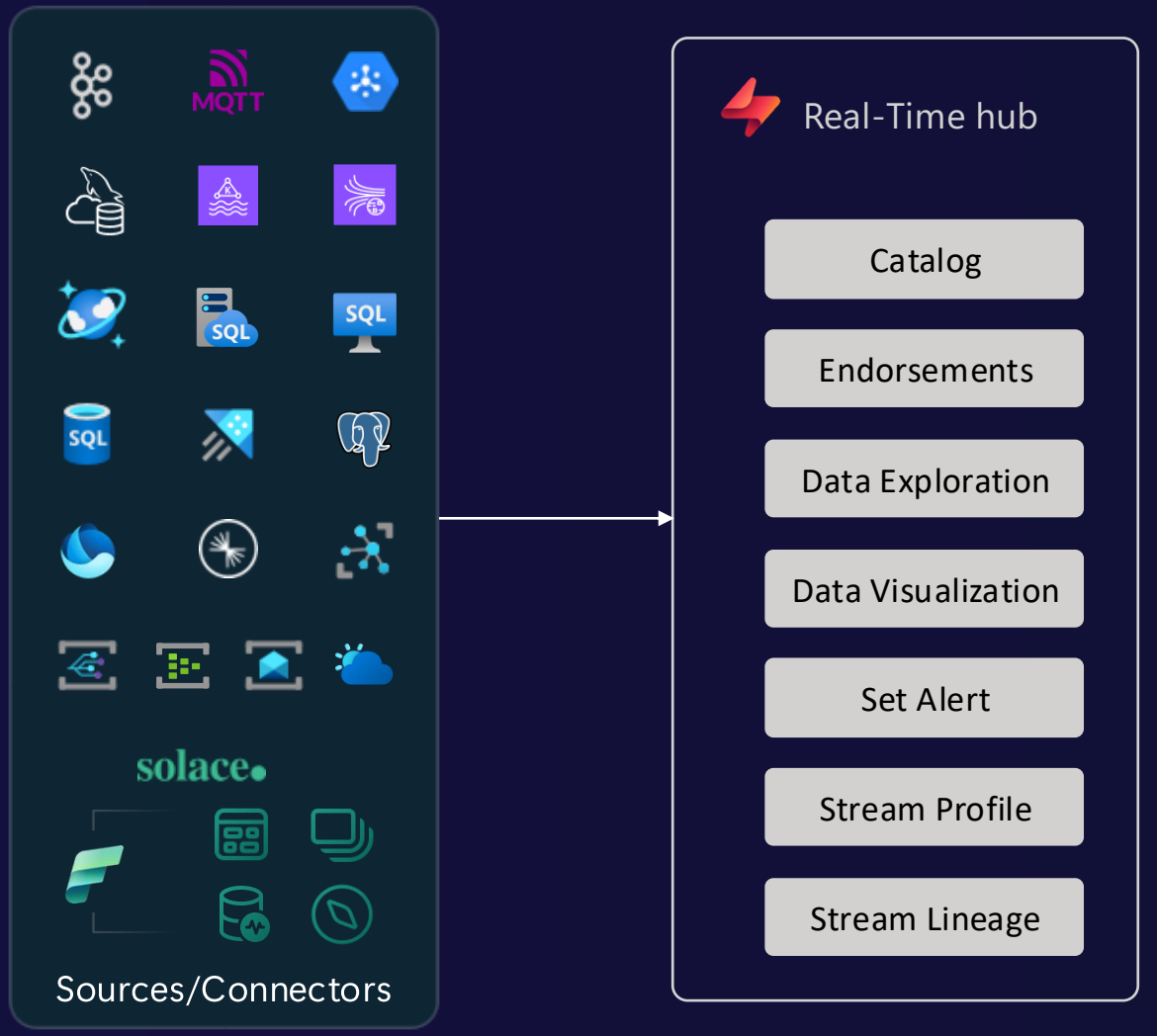
- evnt-user-<XXX>@fabconbar26.onmicrosoft.com
- P: FabCon2026

Head over to aka.bi/lab0

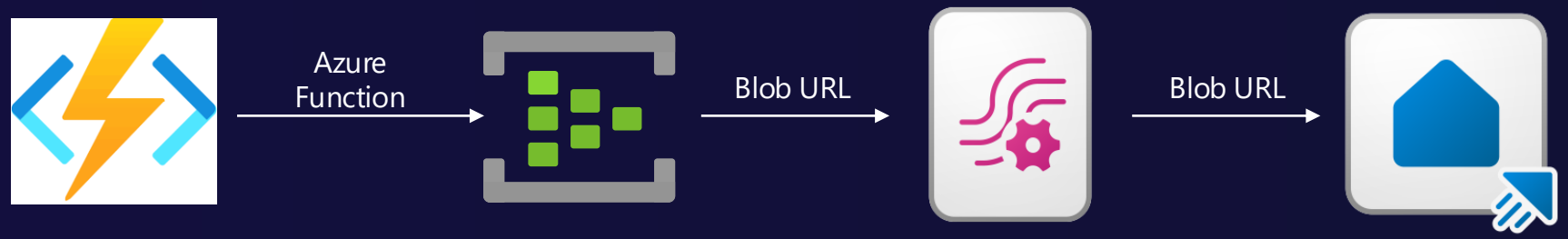
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Real-Time hub



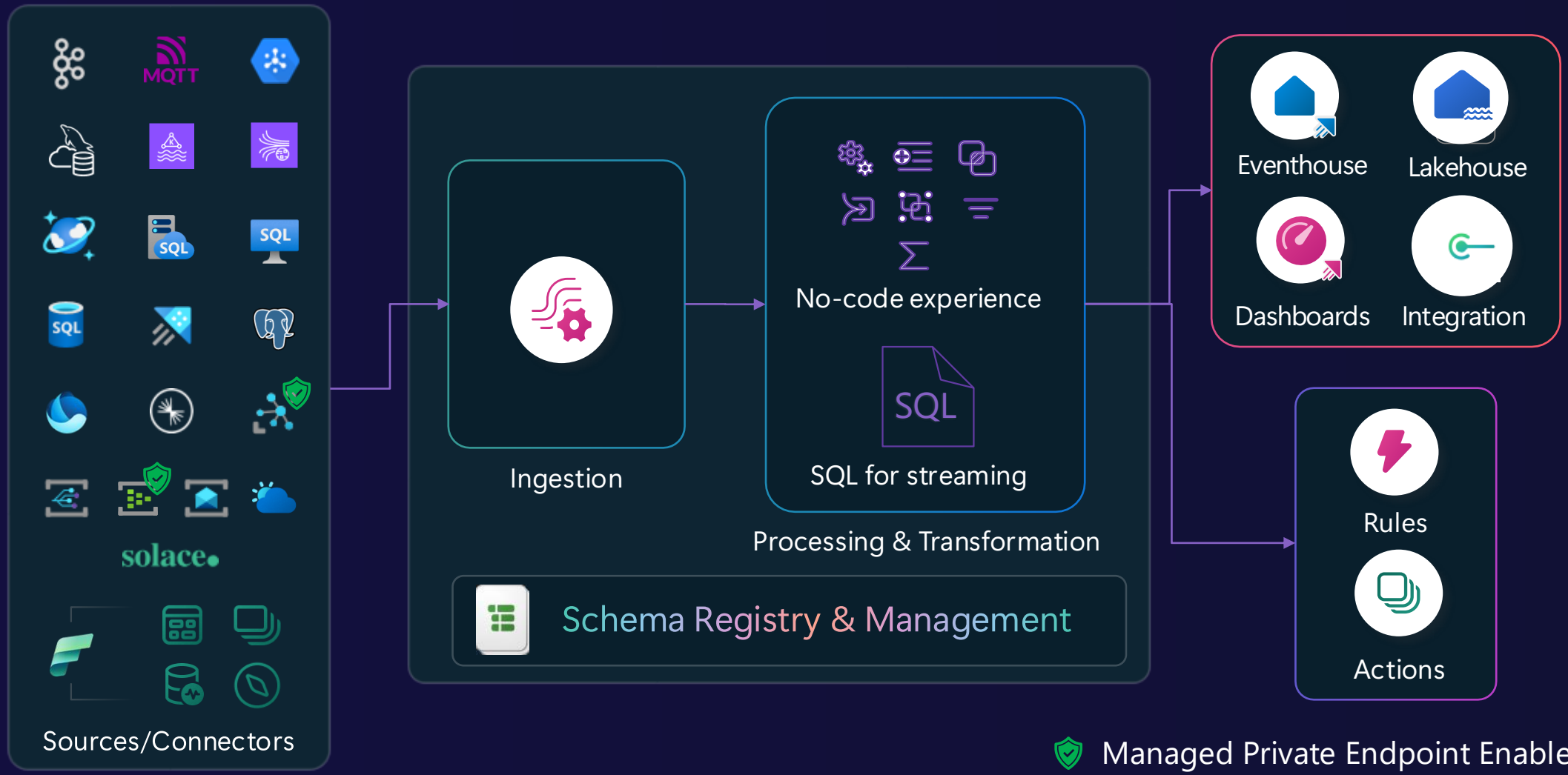
Module 2: Eventstream



1. Read the file
2. Ingest the file

- Instructions follows after anintroduction to Eventstreams

Ingest, process, and route events using Eventstream



Connect events from many types of sources

Examples of connectors and sources

Change Data Capture (CDC)

- CDC for Azure SQL DB
- CDC for Cosmos DB
- CDC for Azure SQL MI
- CDC for Azure PostgreSQL
- CDC for Azure Data Explorer
- CDC for SQL Server
- CDC for MySQL DB
- ...

Message & IoT brokers

- Azure Service Bus
- Solace PubSub+
- Google Pub/Sub
- MQTT protocol
- Azure Event Grid
- Azure IoT Hub
- ...

Streaming brokers

- Apache Kafka
- AWS Kinesis Data Streams
- AWS MSK
- Confluent Cloud
- Azure Event Hubs
- ...

Events

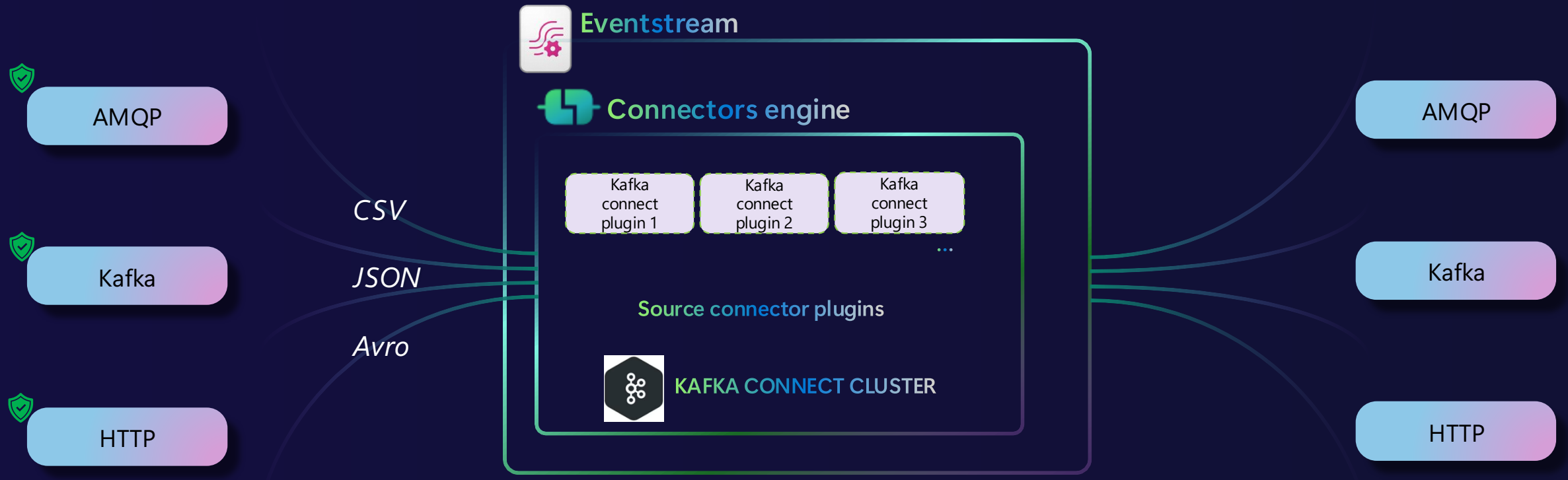
- OneLake events
- Azure Events
- Fabric events
- ...

Feeds

- Weather feed
- ...

Managed Private Endpoint Enabled

Ingest and egress using open protocols and open source



Entra ID Authentication Supported

Eventstream Pricing Model

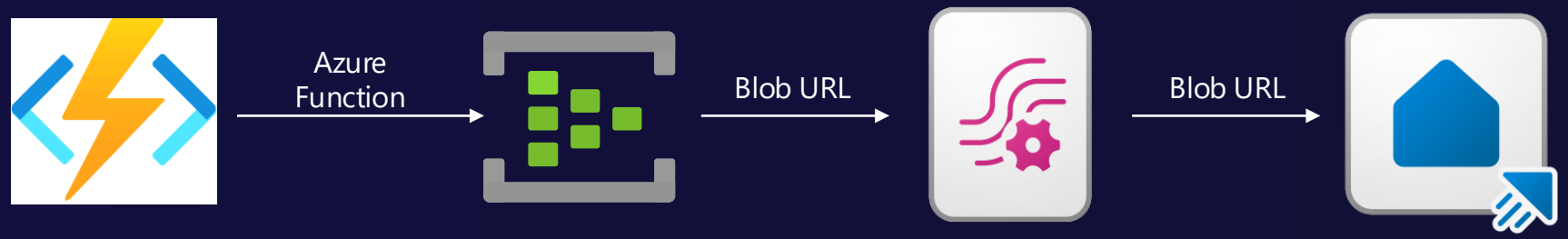
Operation types

Operation	Description	
Eventstream Per Hour	A flat charge while the Eventstream is active (event´s been flowing into the stream). If there's no traffic flowing in or out for the past two hours, no charges apply.	
Eventstream Data Traffic per GB	Data ingress & egress volume in default and derived streams (Includes 24-hour retention)	
Eventstream Processor Per Hour*	Computing resources consumed by the ASA Jobs needed for processing data.	
Eventstream Connectors Per vCore Hour**	Computing resources consumed by the connectors.	

*CU consumption of the Eventstram Processor Per Hour is designed to correlate with throughput, complexity of processing logic, and partition count of input data. The process autoscale accordingly.
**CU consumption of the Eventstream Connectors Per vCore Hour is designed to correlate with throughput. The process autoscale accordingly.

Microsoft Fabric event streams capacity consumption - Microsoft Fabric | Microsoft Learn

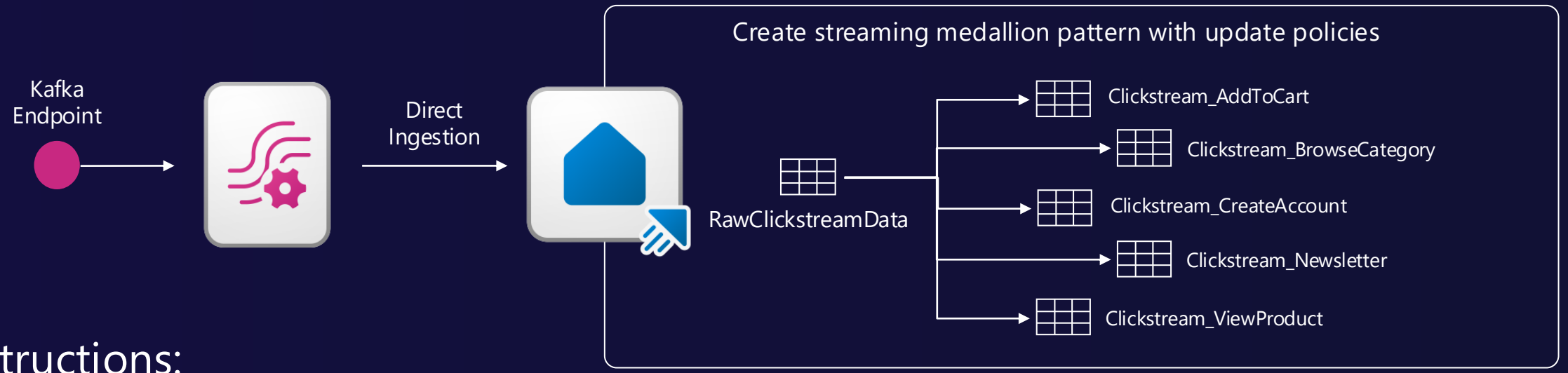
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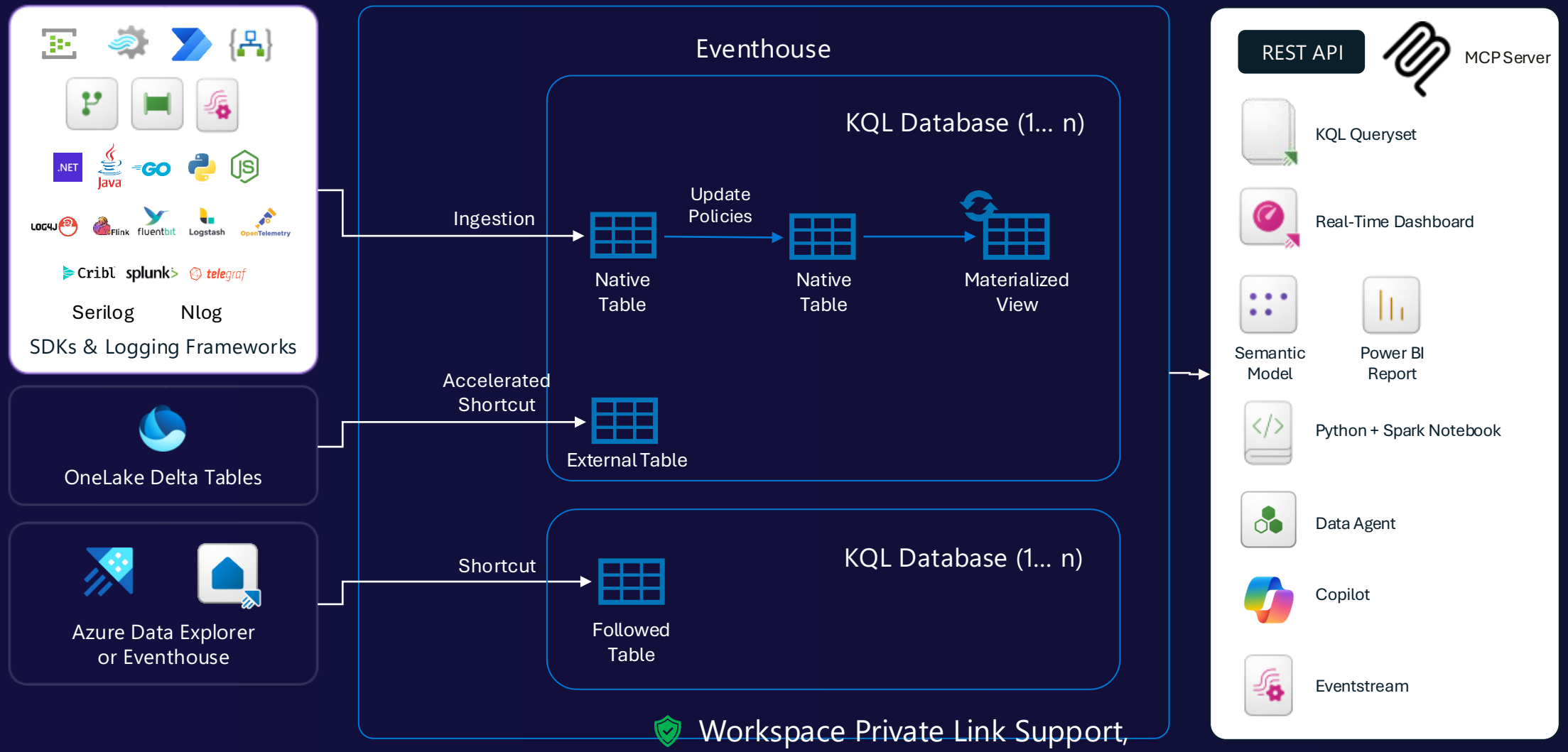
- Instructions:
  - Head over to aka.bi/lab2

Module 3: KQL



- Instructions:
  - First introducing Eventhouse

Analyse and transform events using Eventhouse



Event (Real-Time) Medallion Architecture

Bronze Layer

- Continuously ingest raw data to the raw landing table
- Source can be either streaming or batch ingestion
- Retains the full, unprocessed history

Silver Layer

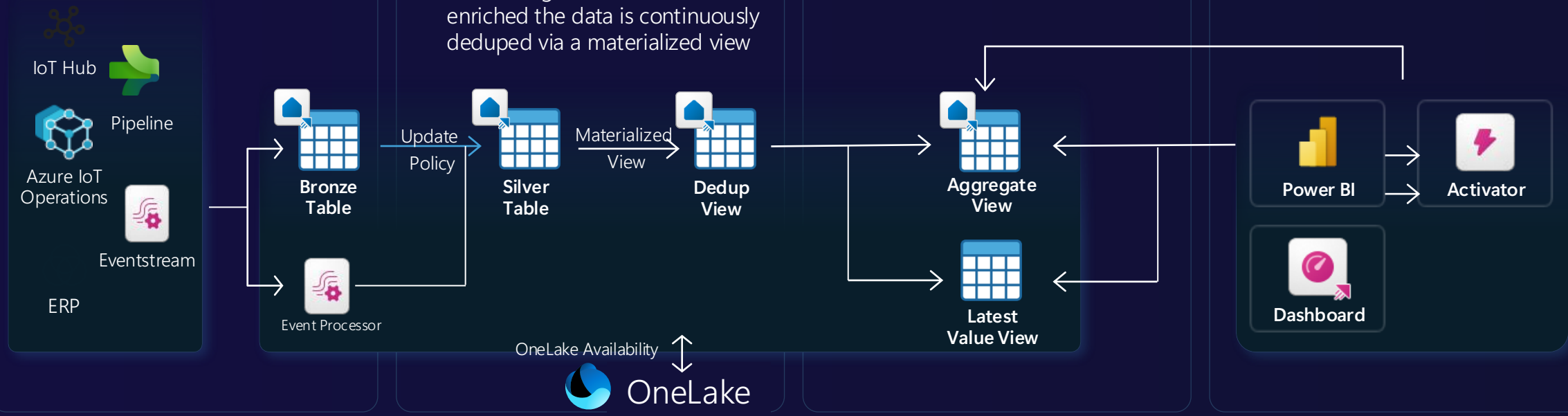
- As data is ingested to the bronze layer it is processed by an update policy or Event Processing in Eventstream and lands into the silver table
- Update policies allow you to transform and enrich the data on a continuous basis
- After being transformed and enriched the data is continuously deduped via a materialized view

Gold Layer

- Continuously aggregate data as it arrives
- Automatically handles late arriving data
- Ability to build multiple gold layer views. For example, hourly aggregates and a latest value view

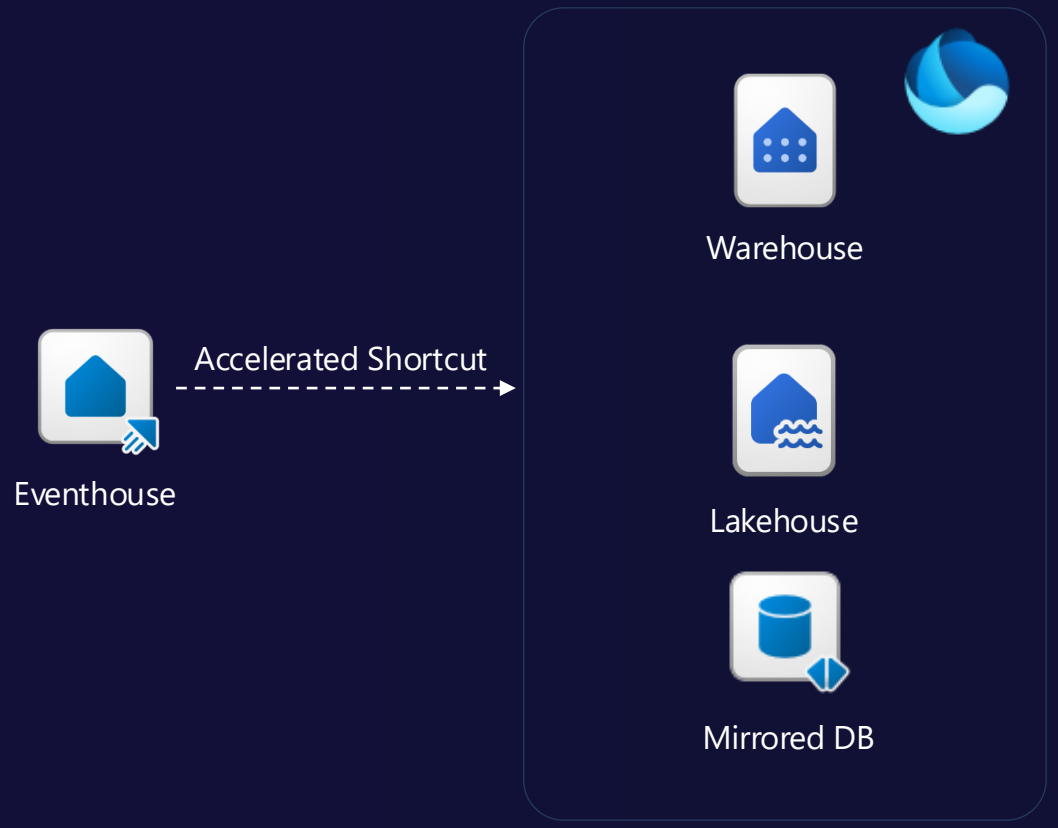
Visualize and activate

- Performance of Eventhouse allows utilize both the deduped view for granular queries and the aggregate view for broader queries
- Reflex allow you to react in real time as both the dedup and aggregation is happening immediately



Query Acceleration

- Query data in OneLake in-place with shortcuts
- Automatic on the fly indexing and caching
- At par performance with ingestion
- Supports Delta as well as Iceberg (with xtable)

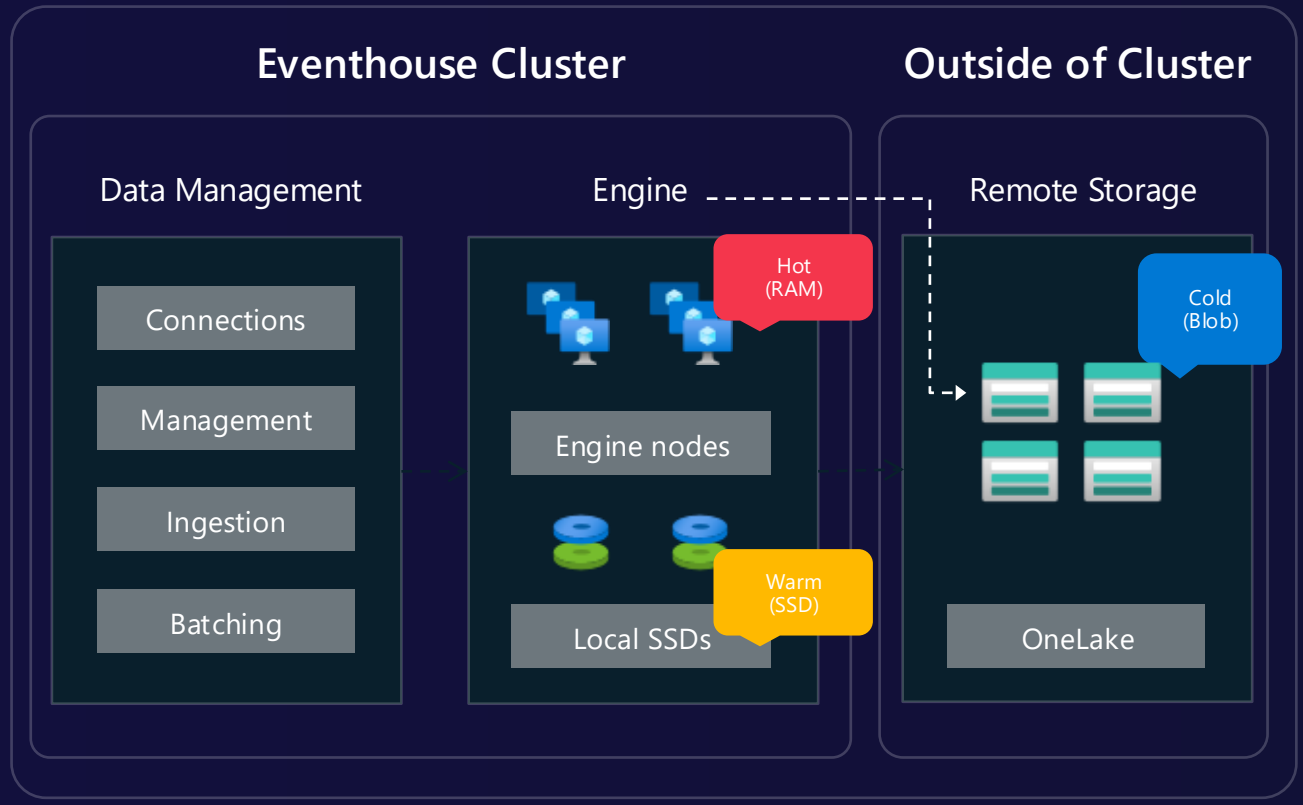


Example:

KQL Command

```
.alter external table MyExternalTable policy query_acceleration
'{"IsEnabled": true, "Hot": "1.00:00:00"}'
```

What´s behind an Eventhouse?



Eventhouse Pricing Model

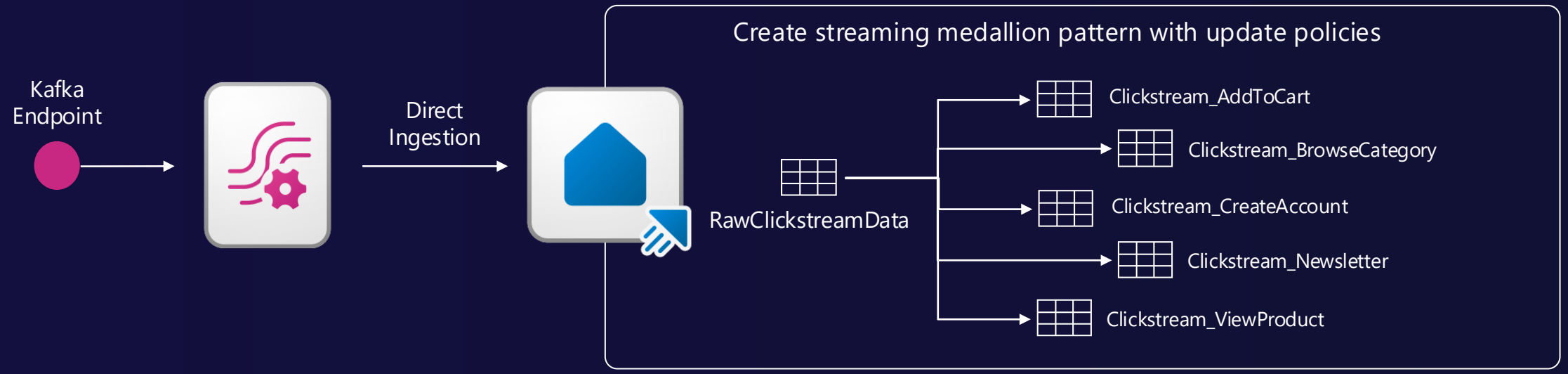
Eventhouse uses a cluster model for the consumption of CU

Pricing	Description	
Eventhouse UpTime	The number of seconds that your eventhouse is active in relation to the number of virtual cores used by your eventhouse. An autoscale mechanism is used to determine the size of your eventhouse. This mechanism ensures cost and performance optimization based on your usage pattern. An eventhouse with multiple KQL databases attached to it only shows Eventhouse UpTime for the eventhouse item.	
OneLake Cache Storage*	Premium storage that is utilized to provide the fastest query response times. When you set the cache policy, you affect this storage tier.*	
OneLake Standard Storage	Standard storage that is used to persist and store all queryable data. When you set the retention policy, you affect this storage tier.	

*Enabling minimum consumption means that you aren't charged for OneLake Cache Storage. When minimum capacity is set, the eventhouse is always active resulting in 100% Eventhouse UpTime.

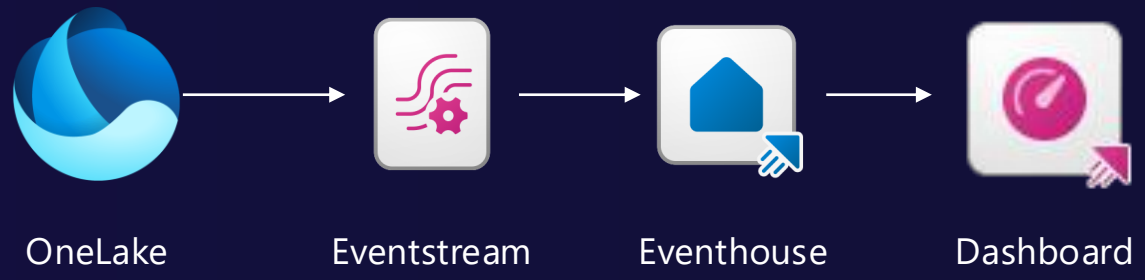
Eventhouse and KQL Database consumption - Microsoft Fabric | Microsoft Learn

Module 3: KQL



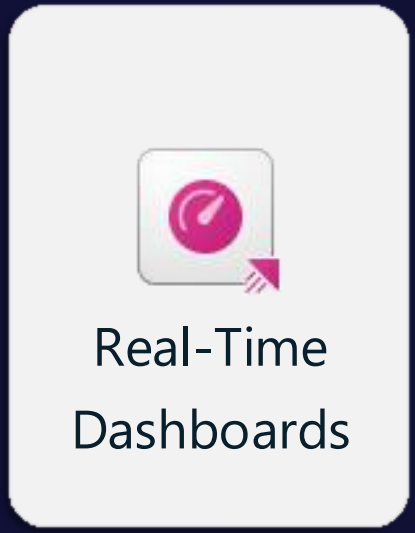
- Instructions:
  - Head over to aka.bi/lab3

Module 4: Real-time Dashboards



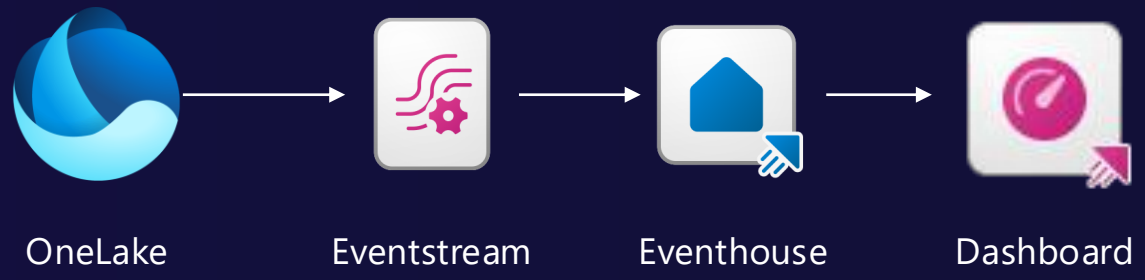
Real-Time Dashboard

Visualizations optimized for streaming, timeseries, event-based data
Complimentary to Power BI
Low-latency refresh
Based on KQL
Parameterized
Cross-interactions and filters



- A Real-Time Dashboard is a workspace item made of **tiles**, each backed by a KQL query against a KQL database (or Azure Data Explorer / Azure Monitor).
- Tiles support every render visual plus dashboard-only ones (map, KPI, multi-stat, table).
- **Parameters** become filters that are pushed *into* the queries, so filtering stays fast.
- **Live refresh** re-runs tiles when new data lands, without reloading the page. Dashboards can be shared, exported as JSON, embedded, and each tile can spawn an **Activator alert**.

Module 4: Real-time Dashboards



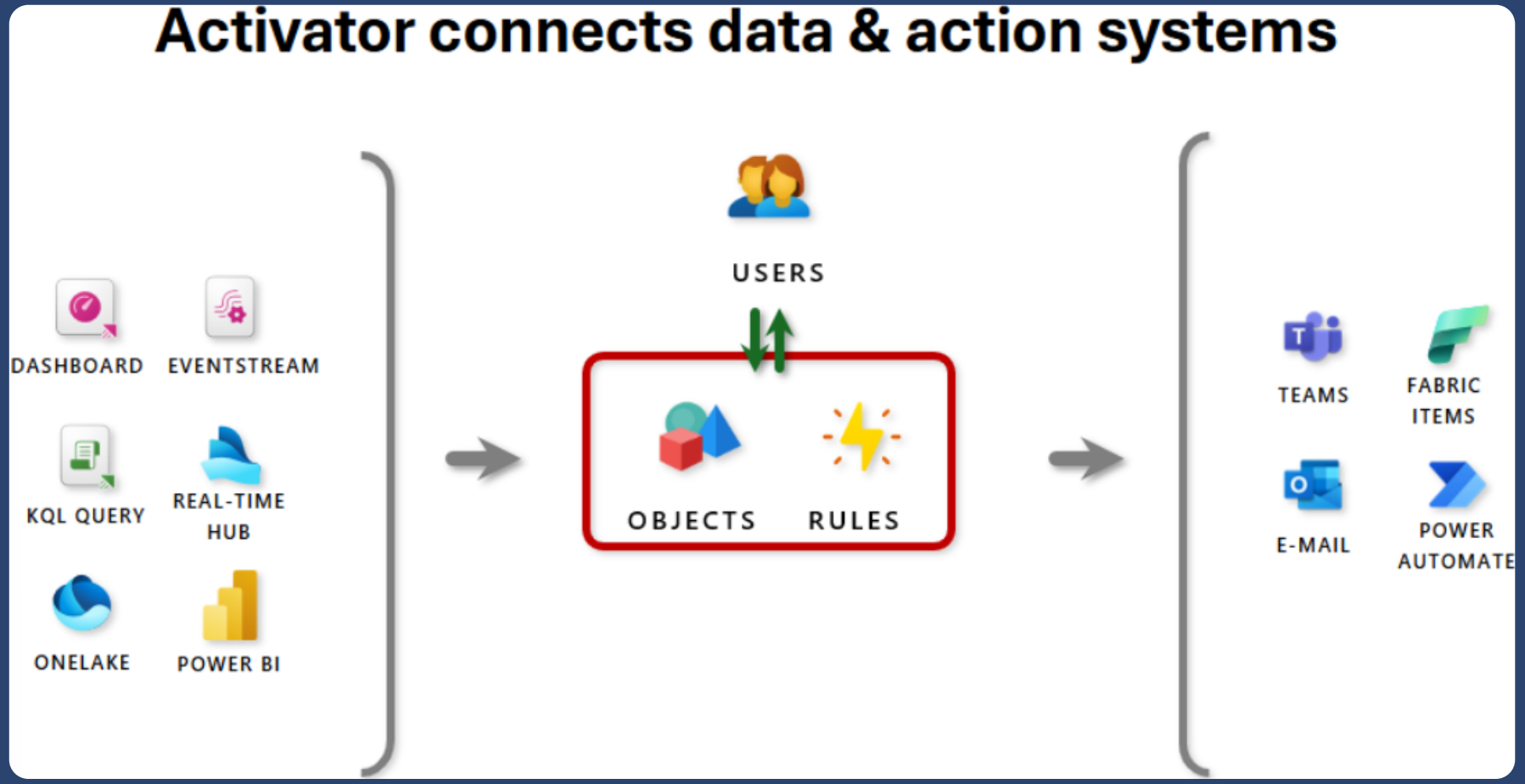
- Instructions:
  - Head over to aka.bi/lab4

Module 5: Activator



- Instructions: A short introduction

Architectural Overview



Core Architectural Elements

Event Sources (Eventstream)



Direct connection to eventstream

Source of events

Subscribe to one or more eventstreams

Fabric events

Activator on Power BI Report or Real-Time Dashboard

Events and Objects



Events

Individual records, rules are evaluated per event

Objects

Events are grouped into objects based on shared identifier [Key].

Object states are monitored over time

Rules and Conditions



Activator has one or more rules; evaluated continously

Actions



Can trigger: Data Pipelines, Power Automate Flows, Notebooks. Notify Emails & Teams

Alert Management and Rules Testing



You can see what might happen based on your current data stream so you can prevent too many actions being triggered

Monitoring and Cost Control



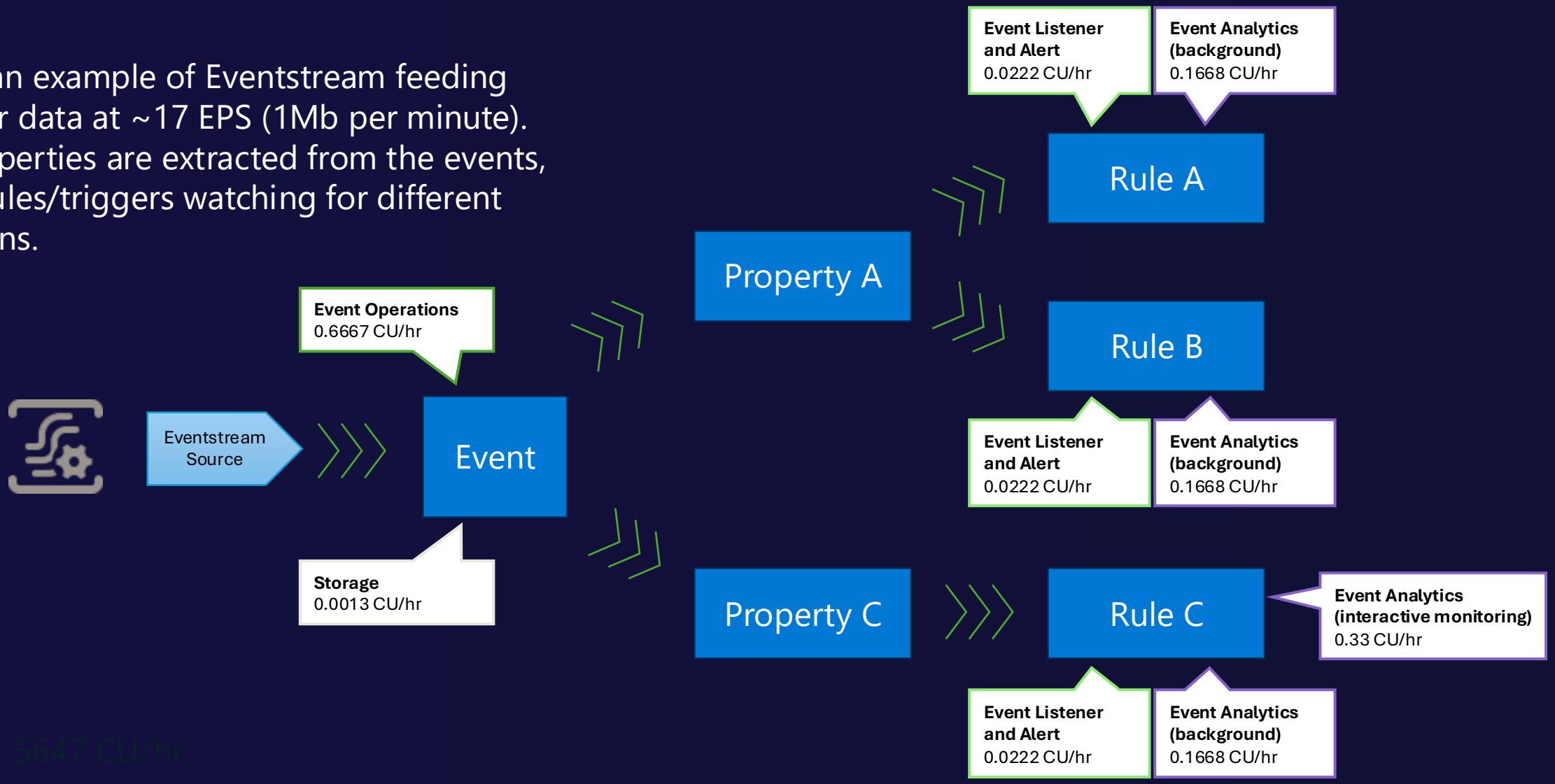
Pricing only applies when rules are executed

Activator – Consumption Meters

BILLING METER	DESCRIPTION	OPERATION UNIT OF MEASURE	FABRIC CONSUMPTION RATE
Real-Time Intelligence – Event Listener & Alert	Flat charge per rule (trigger)	Per hour	0.0222 CU
Real-Time Intelligence – Event Operations	Data ingress	Per M events*	11.111 CU
Data Activator – Event Analytics	Events processed to evaluate rules, run queries etc.	Per M events	2.78 CU
Storage		Per GB/ month	1.333 CU

*Event size 0-1Kb = 1 event, 1-2Kb = 2 events, 2-3Kb = 3 events etc.

Here is an example of Eventstream feeding Activator data at ~17 EPS (1Mb per minute). Two properties are extracted from the events, with 3 rules/triggers watching for different conditions.



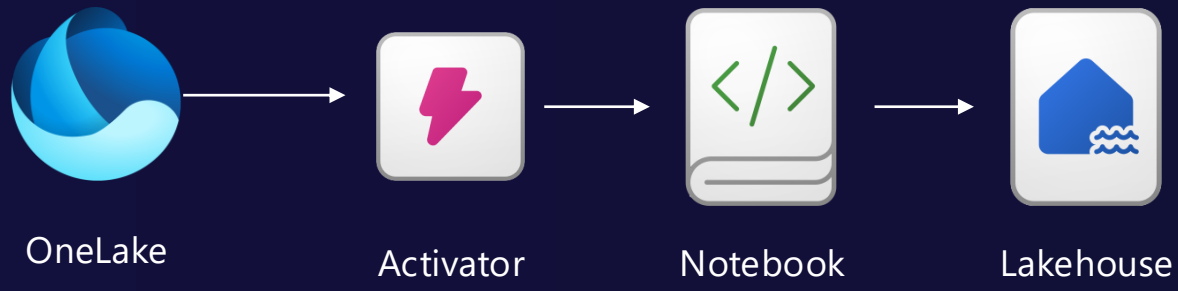
Total = 1.5647 CU/hr

Module 5: Activator



- Instructions:
  - Head over to aka.bi/lab5

Module 6: Fabric Events



- Instructions:
  - Head over to aka.bi/lab6

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